An Accessible Approach to Pricing the European Call by Solving the Black-Scholes PDE

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November 6, 2026

Abstract

In this paper, I aim to derive the fair value of a European call option by solving the Black-Scholes partial differential equation under its terminal (payoff) condition. I first establish the Black-Scholes PDE from a no-arbitrage, continuously hedged position. I then reduce it, through a sequence of changes of variables, to the classical heat equation, whose solution is well known. From that solution I recover the general solution of the Black-Scholes PDE, and I finally specialise it to the European call by imposing the call payoff, together with the terminal condition that, as $\tau = T - t$ tends to 0 (that is, as the contract approaches expiry), the option's value converges to its payoff. This yields the closed-form price of the call.

I would like to sincerely thank Bernard Brooks, mathematician at the Rochester Institute of Technology (Rochester, New York), whose video lectures were invaluable in guiding me through this derivation, and Renaud Bourlès, mathematician at the Department of Applied Mathematics for Finance, École Centrale Marseille. I am likewise indebted to John C. Hull's *Options, Futures, and Other Derivatives*, from which I learned how the Black-Scholes PDE is assembled from a no-arbitrage, continuously hedged portfolio.

1 Assumptions for the Black-Scholes Equation

The Black-Scholes model rests on a set of idealising assumptions: the market is frictionless (no transaction costs or taxes) and permits continuous trading and short selling; the underlying stock pays no dividends over the life of the option; arbitrage opportunities are absent; and both the risk-free rate r and the volatility σ are constant. Under these assumptions, the price of the underlying stock S_t is taken to follow a geometric Brownian motion with constant drift μ and constant volatility σ:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t \tag{1.1}$$

where W_t is a standard Brownian motion. The value B_t of a risk-free bank account then grows deterministically, free of any stochastic component:

$$dB_t = r B_t \, dt \tag{1.2}$$

In full generality, the option price depends on five quantities: the stock price S_t, the time to maturity $\tau = T - t$, the volatility σ, the interest rate r and the strike price K:

$$V = V(\tau,\, S_t;\, r, \sigma, K) \tag{1.3}$$

Since r, σ and K are held constant, the option price reduces to a function of the stock price and the time to maturity alone:

$$V = V(\tau,\, S_t) \tag{1.4}$$

For notational convenience we drop the time subscripts from here on, writing S for S_t, B for B_t and W for W_t. The two dynamics above then read

$$dS = \mu S\, dt + \sigma S\, dW \tag{1.5}$$

and

$$dB = rB\, dt \tag{1.6}$$

2 Derivation of the Black-Scholes PDE

Because V is a smooth function of the stochastic process S and of time, its differential is governed not by ordinary calculus but by Itô's lemma, which introduces a second-order term accounting for the quadratic variation of S:

$$dV = \frac{\partial V}{\partial t}\, dt + \frac{\partial V}{\partial S}\, dS + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\, (dS)^2 \tag{2.1}$$

Substituting the dynamics of the stock price (1.5) gives

$$dV = \frac{\partial V}{\partial t}\, dt + \frac{\partial V}{\partial S}\, (\mu S\, dt + \sigma S\, dW) + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\, (dS)^2 \tag{2.2}$$

To evaluate $(dS)^2$ we expand the square and apply the rules of Itô calculus, namely $(dt)^2 = 0$, $dt\, dW = 0$ and $(dW)^2 = dt$. Only the $\sigma^2 S^2\, (dW)^2$ contribution survives, so that $(dS)^2 = \sigma^2 S^2\, dt$ and

$$dV = \frac{\partial V}{\partial t}\, dt + \frac{\partial V}{\partial S}\, (\mu S\, dt + \sigma S\, dW) + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\, dt \tag{2.3}$$

Expanding the bracket,

$$dV = \frac{\partial V}{\partial t}\, dt + \mu S \frac{\partial V}{\partial S}\, dt + \sigma S \frac{\partial V}{\partial S}\, dW + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\, dt \tag{2.4}$$

and collecting the deterministic (dt) and stochastic (dW) parts yields

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S}\, dW \tag{2.5}$$

We thus obtain a stochastic differential equation with a deterministic drift term and a stochastic diffusion term. The diffusion term carries the same Brownian motion W that drives the underlying stock, and it is precisely this term that exposes the option holder to risk.

The key idea of Black and Scholes is to neutralise this randomness through a *delta-hedging* strategy. We hold the option together with a hedging position made of Δ units of the stock and α units of the bank account:

$$\Pi = \Delta S + \alpha B \tag{2.6}$$

Over an infinitesimal interval, and treating Δ and α as locally constant, the value of this hedging portfolio changes by

$$d\Pi = \Delta\, dS + \alpha\, dB \tag{2.7}$$

Substituting the dynamics (1.5) and (1.2) into (2.7),

$$d\Pi = \Delta\,(\mu S\, dt + \sigma S\, dW) + \alpha\, r B\, dt \tag{2.8}$$

and distributing then grouping the dt terms,

$$d\Pi = (\Delta\,\mu S + \alpha\, r B)\, dt + \Delta\,\sigma S\, dW \tag{2.9}$$

Consider now the combined position $V + \Pi$. Its stochastic part is the sum of the dW terms of (2.5) and (2.9); we choose Δ so that this coefficient vanishes:

$$\sigma S \frac{\partial V}{\partial S} + \Delta\,\sigma S = 0 \tag{2.10}$$

Solving for Δ gives the hedge ratio

$$\Delta = -\frac{\partial V}{\partial S} \tag{2.11}$$

the quantity of stock that renders the combined position instantaneously risk-free.
With this choice the dW terms cancel, and the $\mu S\,\partial V/\partial S$ terms cancel as well, leaving a purely deterministic differential:

$$d(V + \Pi) = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + \alpha\, r B\right) dt \tag{2.12}$$

Since $V + \Pi$ is now riskless, the absence of arbitrage forces it to earn the risk-free rate r over the interval dt:

$$d(V + \Pi) = (V + \Pi)\, r\, dt \tag{2.13}$$

Substituting $\Pi = \Delta S + \alpha B = -\dfrac{\partial V}{\partial S}\, S + \alpha B$,

$$d(V + \Pi) = \left(V - \frac{\partial V}{\partial S}\, S + \alpha B\right) r\, dt \tag{2.14}$$

Equating the two expressions (2.12) and (2.14) for $d(V + \Pi)$ and cancelling the common factor dt,

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + \alpha\, r B = rV - rS\frac{\partial V}{\partial S} + \alpha\, r B \tag{2.15}$$

The $\alpha\, r B$ terms appear identically on both sides and cancel. After rearrangement, we obtain the celebrated Black-Scholes PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0 \tag{2.16}$$

This PDE admits infinitely many solutions, one for each derivative written on the underlying; the one whose derivation is the subject of this paper is the European call option,

$$V(S, t) = S\, N(d_1) - K\, e^{-r(T-t)}\, N(d_2) \tag{2.17}$$

where

$$d_1 = \frac{\ln \frac{S}{K} + \left(r + \frac{1}{2}\sigma^2\right)(T - t)}{\sigma\sqrt{T - t}} \tag{2.18}$$

$$d_2 = \frac{\ln \frac{S}{K} + \left(r - \frac{1}{2}\sigma^2\right)(T - t)}{\sigma\sqrt{T - t}} \tag{2.19}$$

and $N(\cdot)$ denotes the standard normal cumulative distribution function.

3 Reducing the Black-Scholes PDE to the Heat Equation

The Black-Scholes equation (2.16) cannot be integrated directly: its coefficients depend on the stock price S, and it carries the zeroth-order term $-rV$. We therefore simplify it, through a sequence of changes of variables, into the classical heat (diffusion) equation, whose solution is known in closed form. We proceed in three stages: we first reverse time and strip out the discounting, then pass to a logarithmic price variable to make the coefficients constant, and finally remove the remaining first-order term to reach the pure diffusion equation.

The first stage introduces the time to maturity $\tau = T - t$ and seeks a solution of the form:

$$V(S,t) = e^{-r(T-t)}\, U(S,\tau), \qquad \tau = T - t \tag{3.1}$$

Differentiating this expression and substituting it into (2.16) eliminates the discount term and leaves an equation for $U(S,\tau)$ alone.

Differentiating (3.1) term by term, and using the chain rule $\dfrac{\partial U}{\partial t} = -\dfrac{\partial U}{\partial \tau}$ implied by $\tau = T - t$, the three derivatives that enter (2.16) are

$$\frac{\partial V}{\partial t} = e^{-r(T-t)} \left(rU - \frac{\partial U}{\partial \tau} \right) \tag{3.2}$$

$$\frac{\partial V}{\partial S} = e^{-r(T-t)} \frac{\partial U}{\partial S} \tag{3.3}$$

$$\frac{\partial^2 V}{\partial S^2} = e^{-r(T-t)} \frac{\partial^2 U}{\partial S^2} \tag{3.4}$$

Inserting these into (2.16) and factoring out $e^{-r(T-t)}$, which never vanishes, we obtain

$$e^{-r(T-t)} \left[rU - \frac{\partial U}{\partial \tau} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + rS\frac{\partial U}{\partial S} - rU \right] = 0 \tag{3.5}$$

The two rU terms cancel, the discount factor having served its purpose, and we are left with an equation for U alone:

$$\frac{\partial U}{\partial \tau} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + rS\frac{\partial U}{\partial S} \tag{3.6}$$

The discounting is now gone, but the coefficients still depend on S; the next substitution job aims to make the coefficients in front of the partial derivatives constant. To do so, we switch to a logarithmic price variable via the following substitution:

$$S = e^{\xi}, \qquad \xi = \ln S, \qquad \frac{d\xi}{dS} = \frac{1}{S} \tag{3.7}$$

The chain rule then expresses the derivatives with respect to S in terms of derivatives with respect to ξ. For the first derivative,

$$\frac{\partial U}{\partial S} = \frac{\partial U}{\partial \xi}\frac{d\xi}{dS} = \frac{1}{S}\frac{\partial U}{\partial \xi} \tag{3.8}$$

For the second derivative we differentiate (3.8) once more with respect to S; since the factor $1/S$ also depends on S, the product rule applies:

$$\frac{\partial^2 U}{\partial S^2} = \frac{\partial}{\partial S}\left(\frac{1}{S}\frac{\partial U}{\partial \xi}\right) = -\frac{1}{S^2}\frac{\partial U}{\partial \xi} + \frac{1}{S}\frac{\partial}{\partial S}\left(\frac{\partial U}{\partial \xi}\right) \tag{3.9}$$

Applying the chain rule once more to the remaining term, namely $\dfrac{\partial}{\partial S} = \dfrac{1}{S}\dfrac{\partial}{\partial \xi}$, we obtain

$$\frac{\partial^2 U}{\partial S^2} = \frac{1}{S^2}\left(\frac{\partial^2 U}{\partial \xi^2} - \frac{\partial U}{\partial \xi}\right) \tag{3.10}$$

Substituting (3.8) and (3.10) into (3.6), the powers of S cancel and we are left with a constant-coefficient equation:

$$\frac{\partial U}{\partial \tau} = \frac{1}{2}\sigma^2\frac{\partial^2 U}{\partial \xi^2} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial U}{\partial \xi} \tag{3.11}$$

Now what we are looking for is a function $U(\xi, \tau)$ that would fit in (3.11). To do so, we introduce the following substitution:

$$x = \xi + \left(r - \frac{1}{2}\sigma^2\right)\tau \tag{3.12}$$

Hence,

$$U(\xi, \tau) \to U\left(x - \left(r - \frac{1}{2}\sigma^2\right)\tau, \tau\right) \tag{3.13}$$

The new variable contains τ, so when we differentiate we must keep track of which variable is held fixed. With $\dfrac{\partial x}{\partial \xi} = 1$ and $\dfrac{\partial x}{\partial \tau} = r - \dfrac{1}{2}\sigma^2$, the chain rule gives, for the space derivatives,

$$\frac{\partial U}{\partial \xi} = \frac{\partial U}{\partial x}, \qquad \frac{\partial^2 U}{\partial \xi^2} = \frac{\partial^2 U}{\partial x^2}, \tag{3.14}$$

and, for the time derivative,

$$\left.\frac{\partial U}{\partial \tau}\right|_{\xi} = \left.\frac{\partial U}{\partial \tau}\right|_{x} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial U}{\partial x} \tag{3.15}$$

The two time derivatives in (3.15) are *not* the same quantity: on the left τ varies at fixed ξ (the original frame, as in (3.11)); on the right it varies at fixed x (the new traveling frame). The subscripts $|_\xi$ and $|_x$ make this explicit.
Substituting (3.14) and (3.15) into (3.11) produces

$$\left.\frac{\partial U}{\partial \tau}\right|_{x} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial U}{\partial x} = \frac{1}{2}\sigma^2\frac{\partial^2 U}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial U}{\partial x} \tag{3.16}$$

The first-order terms cancel

$$\left.\frac{\partial U}{\partial \tau}\right|_{x} = \frac{1}{2}\sigma^2\frac{\partial^2 U}{\partial x^2} \tag{3.17}$$

Since every variable is now expressed in the (x, τ) frame, the time derivative is unambiguous, and we may drop the subscript. The reduction is complete and we are left with the well-known one-dimensional heat equation:

$$\frac{\partial U}{\partial \tau} = \frac{1}{2}\sigma^2\frac{\partial^2 U}{\partial x^2}, \qquad U = U(x, \tau) \tag{3.18}$$

4 Solving the Heat Equation and Recovering the Call Price

Now that we have an equation that looks very similar to the heat equation in one dimension, we can start to use some of the borrowed mathematics from solving this equation, which can easily be found in any standard physics textbook. The heat equation is among the most thoroughly studied partial differential equations in mathematical physics, where it describes the diffusion of heat. Our equation (3.18) is exactly of this type, with the diffusion coefficient $\frac{1}{2}\sigma^2$ playing the role of the thermal diffusivity factor.
A first important property is that the heat equation conserves the total area under the solution: for any solution decaying at infinity,

$$\int_{-\infty}^{\infty} U(x,\tau)\,dx = \text{constant} \tag{4.1}$$

To solve the equation, we look for its *fundamental solution*: the response to an initial point source concentrated at a point x', from which the solution for an arbitrary initial profile (the option's terminal condition in our case: its payoff) can later be reconstructed. We seek it in the form

$$U(x,\tau) = \tau^{\alpha}\, f(\eta), \qquad \eta = \frac{x-x'}{\tau^{\beta}} \tag{4.2}$$

where α and β are constants to be determined and f depends on the single similarity variable η.
Seeking a solution of this shape is pretty natural, and it carries a decisive practical advantage: substituting (4.2) into the heat equation reduces the two-variable partial differential equation to a single ordinary differential equation for $f(\eta)$, which we know how to solve. The factor τ^{α} in front then carries the time dependence fixed by the conservation law (4.1). We carry out this reduction in the next step.
We now compute the two derivatives entering the heat equation (**??**). Both rely on the chain rule through η, whose partial derivatives are

$$\frac{\partial \eta}{\partial x} = \frac{1}{\tau^{\beta}}, \qquad \frac{\partial \eta}{\partial \tau} = -\beta\,\frac{x-x'}{\tau^{\beta+1}} = -\frac{\beta\,\eta}{\tau}$$

where the last equality uses $x - x' = \eta\,\tau^{\beta}$. We first take the derivatives in x. Since τ^{α} does not depend on x,

$$\frac{\partial U}{\partial x} = \tau^{\alpha}\frac{df}{d\eta}\,\frac{\partial \eta}{\partial x} = \tau^{\alpha-\beta}\frac{df}{d\eta},$$

and differentiating once more in x,

$$\frac{\partial^2 U}{\partial x^2} = \tau^{\alpha-\beta}\frac{d^2 f}{d\eta^2}\,\frac{\partial \eta}{\partial x} = \tau^{\alpha-2\beta}\frac{d^2 f}{d\eta^2} \tag{4.3}$$

Next the derivative in τ. Here both τ^{α} and η depend on τ, so the product and chain rules combine:

$$\begin{aligned}\frac{\partial U}{\partial \tau} &= \alpha\,\tau^{\alpha-1} f(\eta) + \tau^{\alpha}\frac{df}{d\eta}\,\frac{\partial \eta}{\partial \tau} \\ &= \alpha\,\tau^{\alpha-1} f(\eta) - \beta\,\eta\,\tau^{\alpha-1}\frac{df}{d\eta} = \tau^{\alpha-1}\left(\alpha f(\eta) - \beta\,\eta\,\frac{df}{d\eta}\right)\end{aligned} \tag{4.4}$$

Substituting (4.3) and (4.4) into the heat equation (**??**) gives

$$\tau^{\alpha-1}\left(\alpha f - \beta\,\eta\,\frac{df}{d\eta}\right) = \frac{1}{2}\sigma^2\,\tau^{\alpha-2\beta}\frac{d^2 f}{d\eta^2} \tag{4.5}$$

The exponents α and β are still free, and we fix them by requiring that every power of τ cancel, so that (4.5) reduces to an equation in η alone. A first condition comes from the conservation law (4.1): the area under U cannot depend on time. Inserting the ansatz and changing variable with $x - x' = \eta\,\tau^{\beta}$, hence $dx = \tau^{\beta}\,d\eta$,

$$\int_{-\infty}^{\infty} U(x,\tau)\,dx = \int_{-\infty}^{\infty} \tau^{\alpha} f(\eta)\,dx = \tau^{\alpha+\beta}\int_{-\infty}^{\infty} f(\eta)\,d\eta \tag{4.6}$$

Since $\int_{-\infty}^{\infty} f(\eta)\,d\eta$ is a pure number, this is independent of τ only if the prefactor $\tau^{\alpha+\beta}$ is constant, which forces $\alpha = -\beta$. Now substituting $\alpha = -\beta$ into (4.5),

$$-\beta\,\tau^{-\beta-1} f - \beta\,\tau^{-\beta-1}\,\eta\,\frac{df}{d\eta} = \frac{1}{2}\sigma^2\,\tau^{-3\beta}\frac{d^2 f}{d\eta^2} \tag{4.7}$$

and factoring the common term $-\beta\,\tau^{-\beta-1}$ on the left,

$$-\beta\,\tau^{-\beta-1}\left(f + \eta\,\frac{df}{d\eta}\right) = \frac{1}{2}\sigma^2\,\tau^{-3\beta}\frac{d^2 f}{d\eta^2} \tag{4.8}$$

Each side now carries a single power of τ, and they cancel only if those powers are equal:

$$-\beta - 1 = -3\beta \qquad \Longrightarrow \qquad \beta = \tfrac{1}{2}, \qquad \alpha = -\tfrac{1}{2} \tag{4.9}$$

With these exponents, the fundamental solution takes the self-similar form

$$U(x,\tau) = \frac{1}{\sqrt{\tau}}\,f\!\left(\frac{x - x'}{\sqrt{\tau}}\right). \tag{4.10}$$

Setting $\beta = \frac{1}{2}$ in (4.8), the common factor $\tau^{-3/2}$ cancels on both sides, leaving

$$-\frac{1}{2}\left(f + \eta\,\frac{df}{d\eta}\right) = \frac{1}{2}\sigma^2\frac{d^2 f}{d\eta^2} \tag{4.11}$$

Multiplying through by 2 and rearranging, the heat equation is reduced to a second-order ordinary differential equation for f:

$$\sigma^2\frac{d^2 f}{d\eta^2} + \eta\,\frac{df}{d\eta} + f = 0 \tag{4.12}$$

All that is left to do is solving that second order ODE. We solve (4.12) by noticing that its left-hand side is an exact derivative. Consider first the product $\eta f(\eta)$ and differentiate it with respect to η, using the product rule:

$$\frac{d}{d\eta}\big(\eta f(\eta)\big) = \frac{d\eta}{d\eta}\,f(\eta) + \eta\,\frac{df}{d\eta} = f(\eta) + \eta\,\frac{df}{d\eta} \tag{4.13}$$

The middle two terms of (4.12) are therefore nothing but the derivative of $\eta f(\eta)$:

$$\eta\,\frac{df}{d\eta} + f(\eta) = \frac{d}{d\eta}\big(\eta f(\eta)\big) \tag{4.14}$$

Substituting (4.14) into (4.12) gives

$$\sigma^2 \frac{d^2 f}{d\eta^2} + \frac{d}{d\eta}\big(\eta f(\eta)\big) = 0 \tag{4.15}$$

The remaining second-order term is itself a derivative, since $\dfrac{d^2 f}{d\eta^2} = \dfrac{d}{d\eta}\left(\dfrac{df}{d\eta}\right)$, so that

$$\sigma^2 \frac{d}{d\eta}\left(\frac{df}{d\eta}\right) + \frac{d}{d\eta}\big(\eta f(\eta)\big) = 0 \tag{4.16}$$

Both terms now carry the same operator $\dfrac{d}{d\eta}$, which we factor out:

$$\frac{d}{d\eta}\left(\sigma^2 \frac{df}{d\eta} + \eta f(\eta)\right) = 0 \tag{4.17}$$

Writing $\Psi = \sigma^2 \dfrac{df}{d\eta} + \eta f(\eta)$, equation (4.17) states simply that $\dfrac{d\Psi}{d\eta} = 0$. A quantity whose derivative vanishes everywhere is constant, hence

$$\Psi = \sigma^2 \frac{df}{d\eta} + \eta f(\eta) = C_1 \tag{4.18}$$

Recognising the exact derivative has thus lowered the order of the equation from two to one.

The fundamental solution must decay as $|\eta| \to \infty$ in order to remain integrable. In that limit both $\sigma^2\, df/d\eta$ and ηf vanish, which forces $C_1 = 0$. A nonzero C_1 would reintroduce a second, non-decaying branch (involving the imaginary error function), which the decay condition discards. We are left with the separable equation

$$\sigma^2 \frac{df}{d\eta} = -\eta f \tag{4.19}$$

We rewrite (4.19) by dividing both sides by $\sigma^2 f$, which isolates the logarithmic derivative of f on the left:

$$\frac{1}{f}\frac{df}{d\eta} = -\frac{\eta}{\sigma^2} \tag{4.20}$$

We then integrate both sides with respect to η,

$$\int \frac{1}{f}\frac{df}{d\eta}\, d\eta = -\int \frac{\eta}{\sigma^2}\, d\eta \tag{4.21}$$

For the left-hand integral we use the change of variable $u(\eta) = f(\eta)$, whose differential is $du = \dfrac{df}{d\eta}\, d\eta$. The integrand reduces to $\dfrac{du}{u}$, so that

$$\int \frac{1}{f}\frac{df}{d\eta}\, d\eta = \int \frac{du}{u} = \ln u = \ln f \tag{4.22}$$

where we take $f > 0$, since the Gaussian profile we are seeking never vanishes. The right-hand integral is elementary, $-\int \eta/\sigma^2\, d\eta = -\eta^2/(2\sigma^2)$, and therefore

$$\ln f = -\frac{\eta^2}{2\sigma^2} + \text{const.} \tag{4.23}$$

Exponentiating both sides yields the Gaussian similarity profile

$$f(\eta) = A \exp\left(-\frac{\eta^2}{2\sigma^2}\right) \tag{4.24}$$

with $A = e^{\text{const.}}$ a constant of integration.
The constant A is fixed by the conservation law (4.1), normalised so that the fundamental solution carries unit mass, $\int_{-\infty}^{\infty} f(\eta)\, d\eta = 1$. With the Gaussian integral $\int_{-\infty}^{\infty} e^{-\eta^2/(2\sigma^2)}\, d\eta = \sigma\sqrt{2\pi}$,

$$A\,\sigma\sqrt{2\pi} = 1 \qquad \Longrightarrow \qquad A = \frac{1}{\sigma\sqrt{2\pi}} \tag{4.25}$$

With A now fixed, the similarity profile (4.24) becomes the normalised Gaussian

$$f(\eta) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{\eta^2}{2\sigma^2}} \tag{4.26}$$

We now reverse, one by one, the changes of variables that led us here, in order to express the solution back in the original (S, t) variables.
First we restore U. Recalling from (4.10) that $U(x,\tau) = \tau^{-1/2} f(\eta)$ with $\eta = (x - x')/\sqrt{\tau}$, we insert the normalised profile (4.26):

$$U(x,\tau) = \frac{1}{\sqrt{\tau}}\,\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}\left(\frac{x-x'}{\sqrt{\tau}}\right)^2} \tag{4.27}$$

Next we undo the time reversal. Since $\tau = T - t$ (3.1),

$$U(x,t) = \frac{1}{\sqrt{T-t}}\,\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}\left(\frac{x-x'}{\sqrt{T-t}}\right)^2} \tag{4.28}$$

It remains to undo the travelling-frame and log-price substitutions, which together define x. From (3.12) we have $x = \xi + \left(r - \frac{1}{2}\sigma^2\right)\tau$, and from (3.7) that $\xi = \ln S$; with $\tau = T - t$ this gives

$$x = \xi + \left(r - \frac{1}{2}\sigma^2\right)\tau = \ln S + \left(r - \frac{1}{2}\sigma^2\right)(T - t) \tag{4.29}$$

Substituting (4.29) into the numerator $x - x'$ of (4.28), the fundamental solution, expressed in the original variables, reads

$$U(S,t) = \frac{1}{\sqrt{T-t}}\,\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(\ln S + \left(r - \frac{1}{2}\sigma^2\right)(T-t) - x'\right)^2}{2\sigma^2(T-t)}} \tag{4.30}$$

Recalling the discounting substitution (3.1), the option value follows by multiplying through by the discount factor $e^{-r(T-t)}$:

$$V(S,t) = e^{-r(T-t)}\,\frac{1}{\sqrt{T-t}}\,\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(\ln S + \left(r - \frac{1}{2}\sigma^2\right)(T-t) - x'\right)^2}{2\sigma^2(T-t)}} \tag{4.31}$$

This expression solves the Black-Scholes equation, yet it is not the price of a call: it is the response to a single point source located at x', and on its own it cannot reproduce the

call's payoff at maturity. To see why, recall the terminal condition we must ultimately enforce, that the option value collapses onto its payoff as the contract approaches expiry,

$$\lim_{t\to T} V(S,t) = \text{payoff}(S) \tag{4.32}$$

As $t \to T$ the time to maturity $T - t \to 0$, and the Gaussian in (4.31) becomes infinitely tall and infinitely narrow: all of its mass piles up at the single point $x = x'$, while the discount factor tends to 1. The limiting object is the *Dirac delta* $\delta(x - x')$, an idealised spike that is zero everywhere except at $x = x'$, infinitely high there, and encloses unit total area. It is best understood not through its pointwise values but through its behaviour under an integral, the *sifting property*: integrated against any function g, it returns the value of g at the location of the spike,

$$\int_{-\infty}^{\infty} g(x')\,\delta(x - x')\,dx' = g(x) \tag{4.33}$$

A single fundamental solution can therefore reproduce a payoff concentrated at one point only; a realistic payoff, spread over all possible terminal stock prices, cannot be captured by one such term.

The way out rests on linearity. The Black-Scholes equation (2.16) is linear, so if a function solves it, so does any constant multiple, and so does any sum of solutions. We may thus scale (4.31) by an arbitrary constant C,

$$C\,e^{-r(T-t)}\,\frac{1}{\sqrt{T-t}}\,\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{\left(\ln S+\left(r-\frac{1}{2}\sigma^2\right)(T-t)-x'\right)^2}{2\sigma^2(T-t)}} \tag{4.34}$$

and, more generally, superpose fundamental solutions centred at different source points x_i' with coefficients C_i,

$$\sum_{i=1}^{n} C_i\,e^{-r(T-t)}\,\frac{1}{\sqrt{T-t}}\,\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{\left(\ln S+\left(r-\frac{1}{2}\sigma^2\right)(T-t)-x_i'\right)^2}{2\sigma^2(T-t)}} \tag{4.35}$$

which is again a solution. Letting the source points fill the real line and the weights vary continuously, $C_i \to C(x')$, the sum passes to an integral over all source locations:

$$V(S,t) = \int_{-\infty}^{\infty} C(x')\,e^{-r(T-t)}\,\frac{1}{\sqrt{T-t}}\,\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{\left(\ln S+\left(r-\frac{1}{2}\sigma^2\right)(T-t)-x'\right)^2}{2\sigma^2(T-t)}}dx' \tag{4.36}$$

This superposition is the general solution, and the weight $C(x')$ is still free. We fix it with the terminal condition (4.32). Taking $t \to T$ inside (4.36), the kernel collapses to the Dirac delta $\delta(x - x')$, and by the *sifting property* (4.33) the integral simply returns the weight at x, $\lim_{t\to T} V(S,t) = C(x)$. Since this limit must equal the payoff, we identify the weight with the payoff itself, $C(x') = \text{payoff}(S')$.

It is then natural to express the source variable through the same logarithmic change as before, $x' = \ln S'$, so that $dx' = dS'/S'$. As the integration now runs over a stock price, which cannot be negative, the lower limit becomes 0. Using $\ln S - \ln S' = \ln(S/S')$, the solution reads

$$V(S,t) = \int_{0}^{\infty} \text{payoff}(S')\,e^{-r(T-t)}\,\frac{1}{\sqrt{T-t}}\,\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{\left(\ln\frac{S}{S'}+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\frac{dS'}{S'} \tag{4.37}$$

Finally, since the factor $e^{-r(T-t)}\,\frac{1}{\sqrt{T-t}}\,\frac{1}{\sigma\sqrt{2\pi}}$ does not depend on S', we pull it outside the integral:

$$V(S,t) = e^{-r(T-t)}\,\frac{1}{\sqrt{T-t}}\,\frac{1}{\sigma\sqrt{2\pi}}\int_0^\infty \text{payoff}(S')e^{-\frac{\left(\ln\frac{S}{S'}+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\,\frac{dS'}{S'} \tag{4.38}$$

Equation (4.38) is the general solution of the Black-Scholes PDE. In the next section we specialise this payoff to that of the European call, $\max(S'-K,\,0)$, and evaluate the resulting integral, not by direct integration but through a sequence of changes of variables, to recover the celebrated Black-Scholes call price.
This final step will deliver the closed-form formula in terms of the standard normal distribution, the familiar $N(d_1)$ and $N(d_2)$ terms.

5 From the General Solution to the Call Price

To lighten the notation, we gather all the factors that are independent of the integration variable into a single fraction in front of the integral:

$$V(S,t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\int_0^\infty \text{payoff}(S')\,e^{-\frac{\left(\ln\frac{S}{S'}+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\,\frac{dS'}{S'} \tag{5.1}$$

We now specialise the payoff to that of the European call, $\max(S'-K,\,0)$, where K is the strike price:

$$V(S,t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\int_0^\infty \max(S'-K,\,0)\,e^{-\frac{\left(\ln\frac{S}{S'}+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\,\frac{dS'}{S'} \tag{5.2}$$

The call payoff vanishes whenever $S' < K$ and equals $S'-K$ otherwise, so the integrand contributes nothing below the strike. We may therefore restrict the domain of integration to $S' \geq K$, where $\max(S'-K,\,0) = S'-K$:

$$V(S,t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\int_K^\infty (S'-K)\,e^{-\frac{\left(\ln\frac{S}{S'}+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\,\frac{dS'}{S'} \tag{5.3}$$

Distributing the exponential across the bracket $S'-K$ separates the integrand into two parts. In the first, the S' from the payoff cancels the $1/S'$ of the measure, while in the second a factor $1/S'$ remains. The price therefore splits into two integrals,

$$\begin{aligned} V(S,t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\Bigg[&\int_K^\infty e^{-\frac{\left(\ln\frac{S}{S'}+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\,dS' \\ &- K\int_K^\infty e^{-\frac{\left(\ln\frac{S}{S'}+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\,\frac{dS'}{S'}\Bigg] \end{aligned} \tag{5.4}$$

To simplify all that and see clearly what are the two integrals we are going to be integrating, let's consider

$$V(S,t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\,(I_1 - K\,I_2) \tag{5.5}$$

where

$$I_1 = \int_K^\infty e^{-\frac{\left(\ln\frac{S}{S'}+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\, dS' \tag{5.6}$$

$$I_2 = \int_K^\infty e^{-\frac{\left(\ln\frac{S}{S'}+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\, \frac{dS'}{S'} \tag{5.7}$$

The reason why we have separated (5.3) into two pieces is because we are going to use different substitutions in order to solve I_1 and I_2.

5.1 Integrating I_2

As stated earlier, we will not evaluate this integral by brute force. Instead, we carry out a sequence of substitutions and rearrangements designed to cast it into the form of a standard normal distribution evaluated at some upper limit d_2, whose precise expression will emerge along the way. Concretely, our target is to bring I_2 into the shape of

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{d_2} e^{-\frac{\phi^2}{2}}\, d\phi \tag{5.8}$$

the integral of the standard normal density from $-\infty$ up to d_2. The same strategy, applied to I_1 in the next step, will produce the companion term with limit d_1.

The first substitution is $x' = \ln S'$, so that $\dfrac{dx'}{dS'} = \dfrac{1}{S'}$, that is $\dfrac{dS'}{S'} = dx'$. The measure $\frac{dS'}{S'}$ therefore becomes simply dx'; writing $\ln\frac{S}{S'} = \ln S - x'$ and noting that the lower limit $S' = K$ becomes $x' = \ln K$, we obtain

$$I_2 = \int_{\ln K}^\infty e^{-\frac{\left(\ln S - x' + \left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\, dx' \tag{5.9}$$

We now seek a second substitution that recasts the exponent into the standard normal form $-\frac{\phi^2}{2}$. Imposing

$$-\frac{\left(\ln S - x' + \left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)} = -\frac{\phi^2}{2} \tag{5.10}$$

both sides carry a common factor $-\frac{1}{2}$. Cancelling it, equivalently multiplying through by -2, clears the sign and the factor 2 in the denominator, leaving

$$\frac{\left(\ln S - x' + \left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{\sigma^2(T-t)} = \phi^2 \tag{5.11}$$

Taking the positive square root of both sides and solving for ϕ yields our substitution:

$$\phi = \frac{\ln S - x' + \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \tag{5.12}$$

Before applying the substitution, it is worth seeing what the change of variables does to the integral as a whole. The factor relating dx' and $d\phi$ is constant in x', so it will come

out of the integral and multiply the prefactor $\frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}$ of (5.4). For that product to collapse to the standard normal normalisation $\frac{1}{\sqrt{2\pi}}$, we need it to be positive.
Differentiating ϕ as defined in (5.12) would give $\frac{d\phi}{dx'} = -\frac{1}{\sigma\sqrt{T-t}}$, introducing an unwanted minus sign that would have to be carried through the rest of the calculation. Since the numerator of ϕ enters the exponent only through its square, its overall sign is immaterial there, so we are free to reverse its sign. We therefore adopt

$$\phi = \frac{x' - \ln S - \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \tag{5.13}$$

which leaves the exponent $-\frac{\phi^2}{2}$ unchanged but flips the sign of the derivative:

$$\frac{d\phi}{dx'} = \frac{1}{\sigma\sqrt{T-t}}, \qquad \text{hence} \qquad dx' = \sigma\sqrt{T-t}\, d\phi \tag{5.14}$$

The positive factor $\sigma\sqrt{T-t}$ is exactly what will cancel against the $\sigma\sqrt{2\pi(T-t)}$ in the denominator of the prefactor, leaving the clean $\frac{1}{\sqrt{2\pi}}$ of the standard normal density. After applying the substitution into the integral, we are left with:

$$I_2 = \sigma\sqrt{T-t}\int_{\frac{\ln\frac{K}{S} - \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}}^{\infty} e^{-\frac{\phi^2}{2}}\, d\phi \tag{5.15}$$

We are done transforming I_2. Substituting (5.15) back into (5.4) gives

$$\begin{aligned} V(S,t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\Bigg[&\int_K^{\infty} e^{-\frac{\left(\ln\frac{S}{S'} + \left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\, dS' \\ &- K\,\sigma\sqrt{T-t}\int_{\frac{\ln\frac{K}{S} - \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}}^{\infty} e^{-\frac{\phi^2}{2}}\, d\phi\Bigg] \end{aligned} \tag{5.16}$$

The prefactor multiplies both terms of the bracket. Distributing it, the second term acquires the coefficient

$$\frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\, K\,\sigma\sqrt{T-t} = Ke^{-r(T-t)}\,\frac{\sigma\sqrt{T-t}}{\sqrt{2\pi}\,\sigma\sqrt{T-t}} = Ke^{-r(T-t)}\frac{1}{\sqrt{2\pi}}$$

where we have written $\sigma\sqrt{2\pi(T-t)} = \sqrt{2\pi}\,\sigma\sqrt{T-t}$ to expose the cancellation: the $\sigma\sqrt{T-t}$ produced by the substitution cancels the one hidden in the denominator, leaving the standard normal constant $\frac{1}{\sqrt{2\pi}}$. The price then reads

$$\begin{aligned} V(S,t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}&\int_K^{\infty} e^{-\frac{\left(\ln\frac{S}{S'} + \left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\, dS' \\ &- Ke^{-r(T-t)}\frac{1}{\sqrt{2\pi}}\int_{\frac{\ln\frac{K}{S} - \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}}^{\infty} e^{-\frac{\phi^2}{2}}\, d\phi \end{aligned} \tag{5.17}$$

The remaining ϕ-integral is an upper tail of the standard normal density. Writing ℓ for its lower limit,

$$\ell = \frac{\ln\frac{K}{S} - \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$$

the upper-tail identity

$$\frac{1}{\sqrt{2\pi}}\int_{\ell}^{\infty} e^{-\phi^2/2}\,d\phi = 1 - N(\ell)$$

turns the second term of (5.17) into $-Ke^{-r(T-t)}\big[\,1 - N(\ell)\,\big]$. By the symmetry of the Gaussian, $1 - N(\ell) = N(-\ell)$, so we define $d_2 = -\ell$:

$$\begin{aligned} d_2 &= -\frac{\ln\frac{K}{S} - \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \\ &= \frac{\ln\frac{S}{K} + \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \end{aligned}$$

the overall sign turning $\ln\frac{K}{S}$ into $\ln\frac{S}{K}$ and flipping the sign of the drift term. With this, $1-N(\ell) = N(d_2)$, and the second term is simply $Ke^{-r(T-t)}N(d_2)$. Hence (5.17) simplifies to

$$V(S,t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\int_{K}^{\infty} e^{-\frac{\left(\ln\frac{S}{S'} + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\,dS' - Ke^{-r(T-t)}N(d_2) \tag{5.18}$$

with

$$d_2 = \frac{\ln\frac{S}{K} + \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \tag{5.19}$$

With I_2 now fully resolved, we can now focus on solving I_1.

5.2 Integrating I_1

Integrating I_1 is more involved than I_2. Before the same final substitution (5.13) can be applied, a good deal of algebra is needed to reshape the integrand into the same form. As before, we substitute $x' = \ln S'$, so that $\frac{dx'}{dS'} = \frac{1}{S'}$. Here, however, the measure is dS' rather than the $\frac{dS'}{S'}$ of I_2, and $dS' = S'\,dx' = e^{x'}\,dx'$. Writing $\ln\frac{S}{S'} = \ln S - x'$ and sending the lower limit $S' = K$ to $x' = \ln K$, we obtain

$$I_1 = \int_{\ln K}^{\infty} e^{-\frac{\left(\ln S - x' + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)}}\,e^{x'}\,dx' \tag{5.20}$$

In I_2 the factor $\frac{1}{S'}$ cancelled the S' coming from the substitution; here no such cancellation occurs, and the leftover $e^{x'}$ must be absorbed into the exponent. Using $e^a e^b = e^{a+b}$,

$$I_1 = \int_{\ln K}^{\infty} e^{-\frac{\left(\ln S - x' + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2}{2\sigma^2(T-t)} + x'}\,dx' \tag{5.21}$$

Placing both terms over the common denominator $2\sigma^2(T-t)$, the added x' contributes $+\,2\sigma^2(T-t)\,x'$ to the numerator:

$$I_1 = \int_{\ln K}^{\infty} e^{\frac{-\left(\ln S - x' + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 + 2\sigma^2(T-t)\,x'}{2\sigma^2(T-t)}}\,dx' \tag{5.22}$$

Our aim is to reshape this exponent into the form $-\frac{(\cdots)^2}{2\sigma^2(T-t)}$, so that the substitution (5.13) applies exactly as for I_2. We work on the exponent alone. Factoring the overall minus to the front of the fraction flips the sign of the linear term:

$$-\frac{\left(\ln S - x' + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 - 2\sigma^2(T-t)\,x'}{2\sigma^2(T-t)} \tag{5.23}$$

Since the bracketed term is squared, its overall sign is immaterial, so, exactly as we did for I_2, we reverse it to place x' first so that when we will apply the ϕ-substitution and differentiate it with respect to x', there won't be a minus appearing:

$$-\frac{\left(x' - \ln S - \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 - 2\sigma^2(T-t)\,x'}{2\sigma^2(T-t)} \tag{5.24}$$

Our aim is now to expand the squared bracketed piece by applying the square. First, let's lump $\ln S - \left(r - \frac{1}{2}\sigma^2\right)(T-t)$ into one piece of algebra so that it is easier to follow. Pulling the minus sign in front of the bracket turns the minus between $\ln S$ and $(r - \frac{1}{2}\sigma^2)(T-t)$ into a plus, and the fraction becomes:

$$-\frac{\left(x' - \left[\ln S + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right]\right)^2 - 2\sigma^2(T-t)\,x'}{2\sigma^2(T-t)} \tag{5.25}$$

Now let

$$A = \ln S + \left(r - \tfrac{1}{2}\sigma^2\right)(T-t),$$

so that the bracket in (5.24) reads $x' - A$ and it is easier to follow. Expanding the square,

$$-\frac{x'^2 - 2Ax' + A^2 - 2\sigma^2(T-t)\,x'}{2\sigma^2(T-t)} \tag{5.26}$$

The two terms linear in x' combine, $-2Ax' - 2\sigma^2(T-t)\,x' = -2\left[A + \sigma^2(T-t)\right]x'$, giving

$$-\frac{x'^2 - 2\left[A + \sigma^2(T-t)\right]x' + A^2}{2\sigma^2(T-t)} \tag{5.27}$$

We now complete the square in x'. To keep the algebra as easy as possible, write $B = A + \sigma^2(T-t)$ for a moment, so the left-hand side reads

$$x'^2 - 2Bx' + A^2 \tag{5.28}$$

Look at the first two terms, $x'^2 - 2Bx'$, and recognise them as the beginning of a perfect square. Recalling that

$$(x' - B)^2 = x'^2 - 2Bx' + B^2 \tag{5.29}$$

we see that $(x'-B)^2$ reproduces $x'^2 - 2Bx'$ exactly, but drags in an extra B^2. Subtracting that B^2 back to preserve the equality,

$$x'^2 - 2Bx' = (x' - B)^2 - B^2 \tag{5.30}$$

and carrying along the $+A^2$ left untouched at the end,

$$x'^2 - 2Bx' + A^2 = (x' - B)^2 - B^2 + A^2 \tag{5.31}$$

Restoring $B = A + \sigma^2(T-t)$ gives the result:

$$x'^2 - 2\left[A + \sigma^2(T-t)\right]x' + A^2 = \left(x' - \left[A + \sigma^2(T-t)\right]\right)^2 - \left[A + \sigma^2(T-t)\right]^2 + A^2 \tag{5.32}$$

so the exponent becomes

$$-\frac{\left(x' - \left[A + \sigma^2(T-t)\right]\right)^2 - \left[A + \sigma^2(T-t)\right]^2 + A^2}{2\sigma^2(T-t)} \tag{5.33}$$

It remains to simplify the constant $A^2 - [A + \sigma^2(T-t)]^2$. By the difference of squares,

$$\begin{aligned} A^2 - \big[A + \sigma^2(T-t)\big]^2 &= \Big(A - \big[A + \sigma^2(T-t)\big]\Big)\Big(A + \big[A + \sigma^2(T-t)\big]\Big) \\ &= \Big(A - A - \sigma^2(T-t)\Big)\Big(A + A + \sigma^2(T-t)\Big) \\ &= -\sigma^2(T-t)\big(2A + \sigma^2(T-t)\big) \end{aligned} \tag{5.34}$$

Substituting $A = \ln S + \big(r - \frac{1}{2}\sigma^2\big)(T-t)$ into the second factor of this product, the $\sigma^2(T-t)$ terms cancel:

$$\begin{aligned} 2A + \sigma^2(T-t) &= 2\ln S + (2r - \sigma^2)(T-t) + \sigma^2(T-t) \\ &= 2\ln S + 2r(T-t) - \sigma^2(T-t) + \sigma^2(T-t) \\ &= 2\big(\ln S + r(T-t)\big) \end{aligned} \tag{5.35}$$

so that $A^2 - [A + \sigma^2(T-t)]^2 = -2\sigma^2(T-t)\big(\ln S + r(T-t)\big)$. The exponent therefore reduces to

$$-\frac{\big(x' - [A + \sigma^2(T-t)]\big)^2 - 2\sigma^2(T-t)\big(\ln S + r(T-t)\big)}{2\sigma^2(T-t)} \tag{5.36}$$

And the exponent is finally in the right shape. Plugging (5.36) into (5.22),

$$I_1 = \int_{\ln K}^{\infty} e^{-\frac{\left(x' - \left[A + \sigma^2(T-t)\right]\right)^2 - 2\sigma^2(T-t)(\ln S + r(T-t))}{2\sigma^2(T-t)}}\, dx' \tag{5.37}$$

The exponent separates into an x'-dependent part and a constant; splitting the exponential accordingly,

$$I_1 = \int_{\ln K}^{\infty} e^{-\frac{\left(x' - \left[A + \sigma^2(T-t)\right]\right)^2}{2\sigma^2(T-t)}}\, e^{\frac{2\sigma^2(T-t)(\ln S + r(T-t))}{2\sigma^2(T-t)}}\, dx' \tag{5.38}$$

In the second factor the $2\sigma^2(T-t)$ cancels, leaving the constant $e^{\ln S + r(T-t)}$, which we take outside the integral:

$$I_1 = e^{\ln S + r(T-t)} \int_{\ln K}^{\infty} e^{-\frac{\left(x' - \left[A + \sigma^2(T-t)\right]\right)^2}{2\sigma^2(T-t)}}\, dx' \tag{5.39}$$

Since $e^{\ln S + r(T-t)} = e^{\ln S}\, e^{r(T-t)} = S\, e^{r(T-t)}$,

$$I_1 = S\, e^{r(T-t)} \int_{\ln K}^{\infty} e^{-\frac{\left(x' - \left[A + \sigma^2(T-t)\right]\right)^2}{2\sigma^2(T-t)}}\, dx' \tag{5.40}$$

The factor $e^{r(T-t)}$ is worth flagging: once I_1 is returned to the price it will cancel the $e^{-r(T-t)}$ sitting out front, which is precisely why the call term emerges undiscounted as $S\,N(d_1)$.

Restoring $A = \ln S + \big(r - \frac{1}{2}\sigma^2\big)(T-t)$, the centre simplifies neatly,

$$A + \sigma^2(T-t) = \ln S + \big(r + \tfrac{1}{2}\sigma^2\big)(T-t)$$

the drift now carrying $r + \frac{1}{2}\sigma^2$ in place of the $r - \frac{1}{2}\sigma^2$ that produced d_2. The integral becomes

$$I_1 = S\, e^{r(T-t)} \int_{\ln K}^{\infty} e^{-\frac{\left(x' - \left[\ln S + \left(r + \frac{1}{2}\sigma^2\right)(T-t)\right]\right)^2}{2\sigma^2(T-t)}}\, dx' \tag{5.41}$$

This is now exactly the shape of the I_2 integrand, so the same substitution applies. Imposing

$$-\frac{\left(x' - \left[\ln S + \left(r + \frac{1}{2}\sigma^2\right)(T-t)\right]\right)^2}{2\sigma^2(T-t)} = -\frac{\phi^2}{2} \tag{5.42}$$

and proceeding exactly as in (5.10)–(5.13), the bracket already has x' first, so no sign reversal is needed and we obtain directly

$$\phi = \frac{x' - \left[\ln S + \left(r + \frac{1}{2}\sigma^2\right)(T-t)\right]}{\sigma\sqrt{T-t}} \tag{5.43}$$

By construction $\frac{d\phi}{dx'} = \frac{1}{\sigma\sqrt{T-t}}$, hence $dx' = \sigma\sqrt{T-t}\,d\phi$. Applying the substitution and transforming the lower limit (the upper limit again going to $+\infty$),

$$I_1 = S\,e^{r(T-t)}\,\sigma\sqrt{T-t}\int_{\frac{\ln\frac{K}{S} - \left(r+\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}}^{\infty} e^{-\frac{\phi^2}{2}}\,d\phi \tag{5.44}$$

Substituting this into (5.18),

$$\begin{aligned} V(S,t) = {} & \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\,S\,e^{r(T-t)}\,\sigma\sqrt{T-t}\int_{\frac{\ln\frac{K}{S} - \left(r+\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}}^{\infty} e^{-\frac{\phi^2}{2}}\,d\phi \\ & - K e^{-r(T-t)} N(d_2). \end{aligned} \tag{5.45}$$

The coefficient of the first term now collapses through two cancellations:

$$\frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}}\,S\,e^{r(T-t)}\,\sigma\sqrt{T-t} = S\,\underbrace{e^{-r(T-t)}e^{r(T-t)}}_{=1}\,\frac{\sigma\sqrt{T-t}}{\sqrt{2\pi}\,\sigma\sqrt{T-t}} = S\frac{1}{\sqrt{2\pi}}$$

the discount $e^{-r(T-t)}$ meeting the $e^{r(T-t)}$ carried out of I_1, and the $\sigma\sqrt{T-t}$ cancelling against the one inside $\sigma\sqrt{2\pi(T-t)} = \sqrt{2\pi}\,\sigma\sqrt{T-t}$, exactly as for I_2. The price then reads

$$V(S,t) = S\frac{1}{\sqrt{2\pi}}\int_{\frac{\ln\frac{K}{S} - \left(r+\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}}^{\infty} e^{-\frac{\phi^2}{2}}\,d\phi - K e^{-r(T-t)} N(d_2) \tag{5.46}$$

As before, the remaining ϕ-integral is an upper tail. Writing ℓ for its lower limit,

$$\ell = \frac{\ln\frac{K}{S} - \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$$

the identity $\frac{1}{\sqrt{2\pi}}\int_{\ell}^{\infty} e^{-\phi^2/2}\,d\phi = 1 - N(\ell)$ turns the first term into $S\,[\,1 - N(\ell)\,]$. By the symmetry $1 - N(\ell) = N(-\ell)$, we define $d_1 = -\ell$:

$$\begin{aligned} d_1 &= -\frac{\ln\frac{K}{S} - \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \\ &= \frac{\ln\frac{S}{K} + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \end{aligned}$$

so that the first term is simply $S\,N(d_1)$. Collecting both contributions, we arrive at the Black–Scholes formula for the European call:

$$V(S,t) = S\,N(d_1) - K\,e^{-r(T-t)}\,N(d_2) \tag{5.47}$$

with

$$d_1 = \frac{\ln \frac{S}{K} + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \tag{5.48}$$

and d_2 as in (5.19). The two differ only by a shift, $d_1 = d_2 + \sigma\sqrt{T-t}$, reflecting the single change of sign on the $\frac{1}{2}\sigma^2$ term in the drift.

6 Graphical Visualisation

The two normal-distribution terms in the call price, $N(d_1)$ and $N(d_2)$, are most easily understood geometrically. The bell-shaped curve in Figure 1 is the standard normal *density* $\varphi(\phi) = \frac{1}{\sqrt{2\pi}} e^{-\phi^2/2}$, whose total area equals 1. The function $N(\cdot)$ is not this curve itself but the *area* beneath it: for any value x, $N(x)$ is the area lying to the left of x,

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\phi^2/2}\, d\phi \tag{6.1}$$

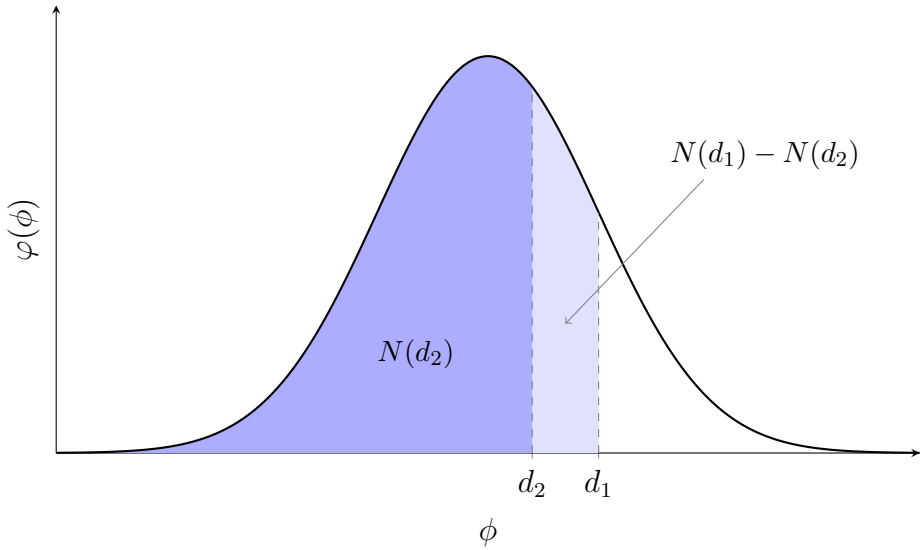


Figure 1: The standard normal density $\varphi(\phi) = \frac{1}{\sqrt{2\pi}} e^{-\phi^2/2}$. The darker area to the left of d_2 is $N(d_2)$; together with the lighter strip it forms the full area to the left of d_1, which is $N(d_1)$.

The darker region is $N(d_2)$, the area to the left of d_2; under the risk-neutral measure this is precisely the probability that the call is exercised. The full shaded region, extending to d_1, is the larger area $N(d_1)$. Since $d_1 = d_2 + \sigma\sqrt{T-t}$ always lies to the *right* of d_2, we have $N(d_1) > N(d_2)$; the thin strip separating them, of width $\sigma\sqrt{T-t}$, is exactly the difference $N(d_1) - N(d_2)$.
When the option is deep in the money ($S \gg K$), both d_1 and d_2 are large and positive: the two vertical lines slide far to the right, almost all of the area is captured, and $N(d_1), N(d_2) \to 1$, so the price approaches its discounted intrinsic value $S - Ke^{-r(T-t)}$. Conversely, deep out of the money ($S \ll K$), both lines slide far to the left, $N(d_1), N(d_2) \to 0$, and the option is worth almost nothing.

7 Summary and Discussion

Starting from the Black–Scholes partial differential equation and reducing it to the heat equation, we have obtained in closed form the fair value at time t of a European call option of strike K and maturity T, written on an underlying of spot price S:

$$V(S,t) = S\,N(d_1) - K\,e^{-r(T-t)}\,N(d_2) \tag{7.1}$$

where

$$d_1 = \frac{\ln\frac{S}{K} + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}, \tag{7.2}$$

$$d_2 = \frac{\ln\frac{S}{K} + \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} = d_1 - \sigma\sqrt{T-t}, \tag{7.3}$$

and $N(\cdot)$ denotes the cumulative distribution function of the standard normal law as it has already been mentioned in Section 2.

Risk-neutral valuation

The fact that the risk-free rate r, and not the stock's real expected return, governs the price above is no accident: it reflects a principle that underlies the whole theory of derivative pricing, known as *risk-neutral valuation*. The principle states that, to value a derivative, one may pretend to live in a *risk-neutral world* – a hypothetical market in which investors are indifferent to risk and therefore demand no additional expected return for bearing it. Our own world is plainly not of this kind: the more risk investors take on, the higher the return they require. Yet, almost miraculously, the value obtained for an option under the risk-neutral assumption is identical to the one it has in the real world. This is what makes the theory workable, as it frees the price from any need to know the risk preferences of buyers and sellers, which are in practice all but impossible to measure.

Additionally, the fact that an option can be priced without reference to risk aversion seems paradoxical at first, since options are themselves risky investments. The resolution is that an option is valued *relative to its underlying*. The risk attitudes of investors are already embedded in the equilibrium price S of the stock: when investors become more risk-averse the stock price falls, but the formula expressing the option price *in terms of* that stock price is left unchanged. The same mechanism is visible in our own derivation. The delta-hedging portfolio was constructed to be instantaneously riskless, and in forming it the stock's real-world drift μ cancelled, leaving a partial differential equation that contains r but not μ. Since the drift is the only place where risk preferences could have entered, the price cannot depend on them.

A risk-neutral world is convenient precisely because two simplifications hold there:

1. the expected return on a stock, or on any other asset, is the risk-free rate r;
2. the rate used to discount expected cash flows, from an option or any other instrument, is again the risk-free rate r.

Under these two rules the call price is nothing but its expected payoff at maturity, computed under the risk-neutral measure $\mathbb{Q}$ and discounted at r,

$$V(S,t) = e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}[\max(S_T - K,\,0)] \tag{7.4}$$

and the two terms of our closed-form result are exactly the two halves of this expectation: $S\,N(d_1)$ is the value of the share received, $K\,e^{-r(T-t)}N(d_2)$ the value of the strike paid, and $N(d_2) = \mathbb{Q}(S_T > K)$ is the risk-neutral probability that the option is exercised.

Reading the formula

Each piece carries a financial meaning. Everything hinges on a single event: the call is exercised only if the stock finishes above the strike, $S_T > K$, and under the risk-neutral measure $N(d_2)$ is precisely the probability of this happening. The second term then reads straight off its factors: the holder pays the strike K on exercise, which occurs with probability $N(d_2)$, so $K\,N(d_2)$ is the expected payment at maturity, and the discount factor $e^{-r(T-t)}$ brings it back to today's money, making $K\,e^{-r(T-t)}\,N(d_2)$ the value today of the strike the holder expects to pay. The first term, $S\,N(d_1)$, is its mirror image on the receiving side, the value today of the share collected at exercise; equivalently, $N(d_1)$ is the option's *delta*, the quantity of the underlying one must hold to hedge the option locally. The price is therefore the value of what the holder receives, the share, minus the value of what they pay, the strike, each scaled by the appropriate probability. The numbers d_1 and d_2 measure, in units of the standard deviation $\sigma\sqrt{T-t}$ of log-returns over the remaining life, how far the option sits in or out of the money, and differ by exactly that one standard deviation, $d_1 = d_2 + \sigma\sqrt{T-t}$.

When the formula applies

The derivation rests on a deliberately idealised market. The option must be *European*, exercisable only at maturity, and written on an underlying that pays *no dividends* over the life of the contract. The price S is assumed to follow a geometric Brownian motion with *constant* volatility σ and *constant* risk-free rate r, so that log-returns are normally distributed. As we have already mentioned in Section 1, the market is taken to be frictionless, no transaction costs or taxes, unrestricted short selling, continuous trading, perfectly divisible assets, and free of arbitrage. Under these hypotheses the delta-hedging argument is exact and the value above is the unique no-arbitrage price.

Limitations and modern extensions

These assumptions are convenient but empirically fragile, and the most fragile is constant volatility. Real markets display a *volatility smile*: options of different strikes and maturities trade at different implied volatilities, which a single constant σ cannot reproduce. Returns also show heavier tails and volatility clustering than a lognormal model permits. For this reason Black–Scholes is today used less as a literal pricing model than as a common language, prices are quoted through their implied volatility, and as a benchmark against which richer models are calibrated.

Those richer models relax the constant-volatility hypothesis by letting the variance itself evolve randomly. The best known is the *Heston* stochastic-volatility model, in which the instantaneous variance v_t follows a mean-reverting square-root diffusion of *Cox–Ingersoll–Ross* type,

$$dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v, \tag{7.5}$$

with κ the speed of mean reversion, θ the long-run variance, and ξ the volatility of volatility. By correlating the variance shocks dW_t^v with the price shocks, this model

reproduces the smile and the asymmetry of observed option prices while remaining semi-analytically tractable. Other common extensions include local-volatility models, which fit the entire observed surface, and jump-diffusion models, which add discontinuous moves to capture sudden crashes. Black–Scholes nevertheless remains the foundation on which all of these are built.